DEFINING WAVEFUNCTION BRANCHES: COHERENCE GRAPH FRAGMENTATION AND THE MONITORING STRUCTURE

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Abstract. The Everettian interpretation of quantum mechanics removes the collapse postulate but inherits an obligation: to define wavefunction branches precisely. We propose a definition resting on five axioms: three from standard quantum mechanics, plus two new ones. First, the monitoring structure of the Hamiltonian — the system-apparatus-environment decomposition and coupling structure — is a necessary physical input. Second, branches are connected components of a coherence graph after environmental decoherence fragments it. The branch partition is identified by the Fiedler eigenvector of the coupling graph Laplacian, a polynomial-time spectral computation with no optimisation or variational freedom.

From these axioms we derive branch formation at an explicit timescale, partition universality (insensitivity to threshold, initial state, and intra-block details within the regime where the monitoring conditions hold), and show that the definition satisfies all four criteria identified by Riedel (2025): forward-in-time tree structure, quasiclassical eigenstates, effective collapse, and bounded entanglement with area-law scaling. We verify the definition against Stern–Gerlach, double-slit, and Bell experiments, and validate it numerically across random Hamiltonian ensembles, random initial states, systematic violations of the monitoring conditions, and multi-branch ($k = 3$) scenarios.

This paper does not attempt to solve the measurement problem. It provides a precise, computable definition of one of its central objects. The monitoring structure is not derived but taken as input; its necessity is demonstrated by showing that branching degrades when the monitoring conditions are violated.

1. Introduction

The Many-Worlds Interpretation of quantum mechanics [2, 3] takes the wavefunction as a complete description of reality and removes the collapse postulate. Under this interpretation, the classical world we experience must correspond to dynamically isolated components

of the quantum state — *branches* — that individually behave quasiclassically. The existence of such branches is central to the Everettian programme, yet no mathematically precise, computable definition has been given.

Decoherence theory [4, 5, 6] explains how quantum systems lose coherence through environmental interaction and identifies the pointer basis as the set of states that survive environmental monitoring. However, decoherence theory describes the *process* by which branches form without defining what a *branch* is: it does not provide a mathematical criterion for when the quantum state has separated into distinct branches, or a procedure for identifying them.

Riedel [9] has crystallised this gap into a concrete challenge. He identifies four criteria that a satisfactory branch definition must meet — tree structure, quasiclassical eigenstates, effective collapse, and bounded entanglement — along with desiderata concerning objectivity [13], uniqueness, and computability. Two recent proposals by Weingarten [10] and Taylor & McCulloch [11], both based on quantum complexity, are evaluated against these criteria and found to face a common obstacle: quantum complexity is generically infeasible to compute.

The central observation behind this paper is that environmental decoherence does not suppress quantum coherence uniformly. It *fragments* the coherence structure of the density matrix. Consider a density matrix $\rho(t)$ evolving under a Hamiltonian H with environmental monitoring. At early times, the off-diagonal elements $\rho_{ij}(t)$ — the coherences[1] — connect every part of the state to every other. As the environment monitors the system, it selectively destroys coherences between states that the environment can distinguish, while preserving coherences between states within the same pointer sector. The result is a structural transition: the matrix of coherences breaks into disconnected blocks.

This transition has a natural graph-theoretic description. Define a graph whose nodes are basis states and whose edges connect pairs with non-negligible coherence. Decoherence takes a connected graph and fragments it into disconnected components (Figure 1). We identify these components as branches. The Hamiltonian defines a second graph — the coupling graph — whose structure predicts which partition will emerge. The branch partition is identified by the Fiedler eigenvector

[1]Throughout this paper, *coherence* between basis states $|i\rangle$ and $|j\rangle$ refers to the off-diagonal density matrix element $\rho_{ij}(t)$. The magnitude $|\rho_{ij}(t)|$ quantifies the capacity of these two states to produce interference.

of the coupling graph Laplacian, a standard tool from spectral graph theory [17], computed in polynomial time.

The definition requires one necessary physical input beyond the Hamiltonian: the monitoring structure — which degrees of freedom constitute the system, apparatus, and environment, and how they couple. This input is not derived from the Hamiltonian operator alone; it is taken as a physical fact about the system. We present numerical evidence that this input is not merely convenient but *necessary*: systematically violating the monitoring conditions causes branching to degrade or fail (Section 5). Every existing approach to the branching problem requires structural input of comparable kind (Section 6). Whether the monitoring structure can be extracted from the Hamiltonian given additional structure such as locality is an open question discussed in Section 7.2.

This paper makes three kinds of claims, each marked explicitly by environment type: Definitions *are mathematical constructions that can be checked for internal consistency but not proved or disproved.* Theorems *and* Propositions *are derived results that follow from the axioms and stated premises; their proofs are given.* Conjectures *are claims supported by numerical evidence but not yet proved; they are labelled as such wherever they appear.*

Outline. Section 2 states the axioms and develops the mathematical definitions. Section 3 derives branch formation from the axioms. Section 4 establishes that the definition satisfies Riedel's four criteria. Section 5 verifies the definition against standard experiments. Section 6 compares with existing proposals. Section 7 discusses scope and limitations.

2. Definitions

2.1. **Axioms.** The definition rests on five axioms. The first three are standard quantum mechanics; the last two are new.

- **A1.** *Hilbert space.* The state of a quantum system is described by a density operator ρ on a Hilbert space $\mathcal{H}$.
- **A2.** *Unitary evolution.* The total state evolves under the von Neumann equation: $i\hbar\, d\rho/dt = [H, \rho]$.
- **A3.** *Tensor product composition.* Composite systems are described on tensor product Hilbert spaces: $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_A \otimes \mathcal{H}_E$.
- **A4.** *Monitoring structure.* The monitoring structure of the Hamiltonian — which degrees of freedom constitute the system, the apparatus, and the environment, and how they couple — is a physical input. Given this structure, the reduced dynamics of

the system-plus-apparatus is obtained by tracing over the environment.

A5. *Branch definition.* Branches are connected components of the coherence graph $G_\rho(t)$ after environmental decoherence has fragmented it. The branch partition is identified by the Fiedler eigenvector of the coupling graph G_H.

Axioms A1–A3 are shared by all interpretations of quantum mechanics. Axiom A4 is the framework's necessary physical input: instead of postulating that measurements produce definite outcomes with Born-rule probabilities, we take the monitoring structure as given and derive everything else. It specifies *which* system is being monitored by *which* environment; whether this input can be derived from the Hamiltonian given additional structure (e.g., locality) is an open question (Section 7.2). Numerical stress tests confirm that violating the monitoring conditions (M1–M3) causes branching to degrade, supporting the claim that A4 is necessary (companion suite, Test 15). Axiom A5 provides the mathematical definition of branches.

Definition 2.1 (Monitoring structure)**.** The *monitoring structure* of a Hamiltonian H_{total} is the specification of: (i) the tensor product decomposition $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_A \otimes \mathcal{H}_E$ identifying the system, apparatus, and environment; and (ii) the coupling structure of H_{total} with respect to this decomposition, including which subsystems interact and through which operators. The monitoring structure is a property of the physical system that the Hamiltonian describes — it is determined by the physical degrees of freedom and their interactions — but it is not recoverable from the operator H_{total} alone without additional structure such as locality or tensor product factorisation assumptions.

The monitoring structure is not a new law of physics but a physical fact about the system: which degrees of freedom exist, how they interact, and which constitute the environment. The branch definition (A5) is a mathematical construction: it defines "branch" in terms of computable properties of the density matrix and the Hamiltonian.

2.2. **The flow current.** In an orthonormal basis $\{|i\rangle\}$, the von Neumann equation [1] gives the diagonal evolution:

$$\frac{d\rho_{ii}}{dt} = \frac{2}{\hbar}\sum_k \operatorname{Im}(H_{ik}\rho_{ki}) = \sum_k J_{k\to i}, \tag{1}$$

where the *flow current* from node k to node i is:

$$J_{k\to i} = \frac{2}{\hbar}\operatorname{Im}(H_{ik}\rho_{ki}). \tag{2}$$

The flow current is antisymmetric ($J_{k\to i} = -J_{i\to k}$) and conserves total weight ($\sum_i d\rho_{ii}/dt = 0$). When $H_{ik} = 0$, the flow current $J_{k\to i}$ vanishes identically regardless of ρ_{ki}: states with no Hamiltonian coupling cannot exchange amplitude.

Remark 2.1 (Experimental access via weak measurement). The flow current $J_{k\to i}$ depends on the off-diagonal density matrix element $\rho_{ki}(t)$, which is not directly observable in a single projective measurement but is in principle accessible through weak-measurement protocols. Repeated weak measurements of the pointer observable S_{op}, each extracting partial information without collapsing the state, can in principle reconstruct the coherence $\rho_{ki}(t)$ between pointer sectors via quantum state tomography of $\rho_{SA}(t)$. In systems where such tomography is feasible (e.g., superconducting qubits, trapped ions), the formation time t^* would correspond to the time at which inter-sector coherences drop below experimental resolution. We note this as a potential experimental signature rather than a near-term prediction: the suppression of inter-branch flow current would manifest as a transition from correlated to independent weak-measurement statistics.

2.3. **The two-layer graph structure.**

Definition 2.2 (Coupling graph)**.** The coupling graph $G_H = (V, E_H, w_H)$ has node set $V = \{|i\rangle\}$, edge set $E_H = \{(i,k) : H_{ik} \neq 0\}$, and edge weights $w_{ik} = |H_{ik}|$.

Definition 2.3 (Coherence graph)**.** The coherence graph $G_\rho(t) = (V, E_\rho(t), w_\rho(t))$ has edge weights $w_{ik}(t) = |\rho_{ik}(t)|$ and node weights $\rho_{ii}(t)$.

G_H encodes which pairs of states can exchange amplitude. $G_\rho(t)$ encodes which pairs currently maintain coherence. The flow current depends on both: $J_{k\to i} \neq 0$ requires $H_{ik} \neq 0$ (edge in G_H) *and* $\rho_{ki} \neq 0$ (edge in $G_\rho(t)$). The physically important case is $H_{ik} \neq 0$ but $\rho_{ki} \approx 0$: coupled but decohered. No flow despite physical coupling. This is the branch-formed state.

2.4. **Branch definition.**

Definition 2.4 (Branch)**.** Let $G_\rho(t)$ be the coherence graph at time t and let $\varepsilon > 0$ be a threshold small relative to the intra-component coherence scale. A *branch* is a connected component of $G_\rho(t)$ at threshold ε: a maximal set of nodes $\mathcal{B} \subseteq V$ such that for any $i, j \in \mathcal{B}$ there exists a path $i = n_0, n_1, \ldots, n_p = j$ with $|\rho_{n_q n_{q+1}}(t)| \geq \varepsilon$ for each step, and for any $i \in \mathcal{B}$, $k \notin \mathcal{B}$: $|\rho_{ik}(t)| < \varepsilon$.

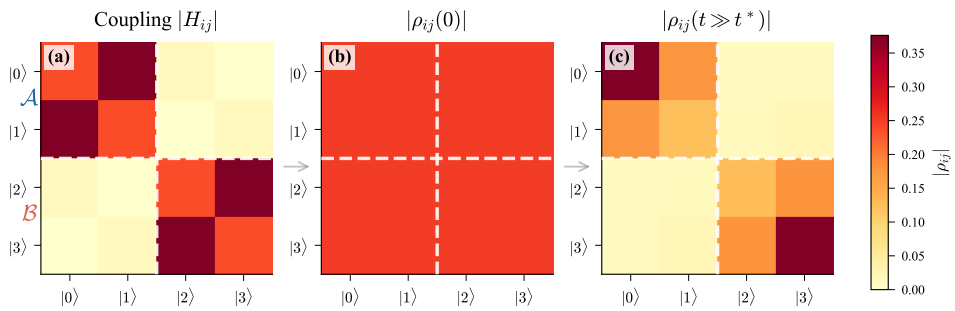

FIGURE 1. Branch formation as matrix fragmentation. (a) The coupling matrix $|H_{ij}|$: two strongly-coupled diagonal blocks ($\mathcal{A}$ and $\mathcal{B}$, identified by the Fiedler eigenvector) connected by weak inter-sector coupling. (b) The density matrix $|\rho_{ij}(0)|$ at $t = 0$: all entries are populated, reflecting full coherence across the entire basis. (c) At late times $t \gg t^*$: the inter-sector blocks (off-diagonal) have decayed to near zero while the intra-sector blocks (diagonal) retain their coherence. The density matrix has fragmented into two independent blocks — the branches. Dashed lines mark the sector boundary. Same physical model as Figure 3 (dimension $4 \times 64 = 256$).

Remark 2.2 (Role of ε and its physical scale). The threshold ε determines *when* branches are considered formed (the formation time t^* depends on ε) but not *which* partition emerges: under selective dephasing, all inter-sector coherences cross any threshold ε before any intra-sector coherence does, yielding the same partition for all ε in the achievable range (Theorem 2.1). The threshold is not a free parameter of the definition; it is a resolution parameter analogous to a regularisation cutoff, and the branch partition is insensitive to its value.

The achievable range of ε is set by the physical energy scales. The lower bound is the residual floor $C/\gamma_{\text{inter}} = 2\|H_{\text{inter}}\|_F/(\hbar\,\gamma_{\text{inter}})$: the ratio of the inter-sector Hamiltonian coupling to the inter-sector dephasing rate. Below this floor, the Hamiltonian regenerates coherence as fast as the environment destroys it, and branches cannot be resolved. The upper bound is the initial inter-sector coherence $x_0 = \|\rho_{\text{inter}}(0)\|_F$. For any ε in $(C/\gamma_{\text{inter}},\, x_0)$, the formation theorem guarantees branch formation. For the primary 4-state model used throughout this paper, $C/\gamma_{\text{inter}} \approx 0.059$ and $x_0 \approx 0.71$, giving an achievable range spanning more than one order of magnitude. In macroscopic systems,

$\gamma_{\text{inter}} \propto J(\omega)$ grows with the number of environment modes while $\|H_{\text{inter}}\|_F$ is fixed, so the floor shrinks and the achievable range widens.

Remark 2.3 (Sectors and branches). We distinguish two related concepts. A *sector* is a partition element of the coupling graph G_H, identified by the Fiedler eigenvector of the graph Laplacian (Section 2.5). Sectors are static properties of the Hamiltonian. A *branch* is a connected component of the coherence graph $G_\rho(t)$ at time t. Branches are dynamical objects that form when decoherence fragments the coherence graph. The central result (Theorem 2.1) is that the branch partition coincides with the sector partition: decoherence fragments $G_\rho(t)$ along exactly the boundaries predicted by the Fiedler analysis of G_H.

The net inter-branch flow current is bounded: for branches $\mathcal{A}$ and $\mathcal{B}$,

$$\left|\left.\frac{dw_{\mathcal{A}}}{dt}\right|_{\text{inter}}\right| \leq \frac{2}{\hbar}\|H_{\text{inter}}\|_F\,\|\rho_{\text{inter}}\|_F. \tag{3}$$

After branch formation ($|\rho_{ij}| < \varepsilon$ for all inter-branch pairs), $\|\rho_{\text{inter}}\|_F < \sqrt{|\mathcal{A}||\mathcal{B}|}\,\varepsilon$, giving a bound that vanishes with ε. Each branch evolves as if the others did not exist, to within this residual.

2.5. **Partition discovery via the Fiedler eigenvector.** The coupling graph G_H is constructed from $|H_{ij}|$ alone — no labelling of system, apparatus, or pointer states is required. The partition of V into sectors is not an input but an *output*: it is the sign partition of the Fiedler eigenvector (the eigenvector for the second-smallest eigenvalue λ_2 of the graph Laplacian $L = D - W$, where $D = \text{diag}(\sum_j |H_{ij}|)$ and $W_{ij} = |H_{ij}|$).

The Fiedler gap λ_2 quantifies how well-separated the communities are:

- When λ_2 is large relative to the intra-community spectral scale, the partition is robust and branches will form cleanly.
- When λ_2 vanishes, the graph has no natural bipartition and no branching occurs.
- The gap serves as a built-in diagnostic for the definition's domain of applicability (Figure 2).

For multi-sector branching ($k > 2$ sectors), two approaches are available. The first is *simultaneous k-cluster spectral partitioning*: embed each node using the first $k-1$ eigenvectors of the graph Laplacian (the eigenspace between $\lambda_1 = 0$ and λ_k), then apply k-means clustering in

this eigenspace. The number of clusters k is detected from the Laplacian spectrum as the multiplicity of near-zero eigenvalues (the eigengap heuristic: choose k to maximise λ_{k+1}/λ_k). This method is standard in spectral graph theory but introduces a variational step (k-means) that the $k = 2$ case avoids.

The second approach is *recursive Fiedler bisection*: repeatedly bipartition the largest remaining community. This preserves the variational-free character of the $k = 2$ case but does not reliably recover the full partition in general, because the Fiedler vector projects a $(k-1)$-dimensional separation onto one dimension and can misassign states at the first cut. However, for the physically relevant case — sequential branching events, each driven by a separate decoherence channel at a different time — each bisection corresponds to a distinct decoherence event, and the projection issue does not arise. The companion suite confirms this for $k = 3$: sequential bisection correctly recovers all three sectors (Test 2 in [23]).

Detecting k from the Laplacian spectrum and rigorously proving correctness of the simultaneous method for general k remain open problems. The diagnostic generalises naturally: the ratio λ_k/λ_{k+1} plays the role of λ_2/λ_3.

Remark 2.4 (Scope of the discovery). The Fiedler eigenvector discovers the *branch partition* — which basis states group into which sectors — without being told which states are pointer states. It does not discover the system-environment split; that is the role of Axiom A4. The definition takes as input an open-system description (a Hamiltonian H_{SA} and Lindblad operators from tracing out the environment); the Fiedler construction then identifies the branch structure within that open system.

2.6. **Partition universality.** The threshold ε determines *when* branches form but not *which* partition emerges.

Theorem 2.1 (Partition universality)**.** Under selective dephasing (Theorem 3.1) and block-structured G_H with $H_{\text{inter}}/H_{\text{intra}} < \theta_c$ (where $\theta_c := \lambda_1^{\text{intra}}/(4\sqrt{2d}\,\Delta^{\text{inter}})$ is the Fiedler block-alignment threshold from the Davis–Kahan perturbation bound), the branch partition is the sign partition of the Fiedler eigenvector of $L(G_H)$, independent of:

(a) the threshold ε (within the achievable range $C/\gamma_{\text{inter}} < \varepsilon < x_0$);
(b) the initial state $\rho(0)$ (within the class of states not already diagonal in the pointer basis);
(c) the fine details of the Hamiltonian matrix elements within each block.

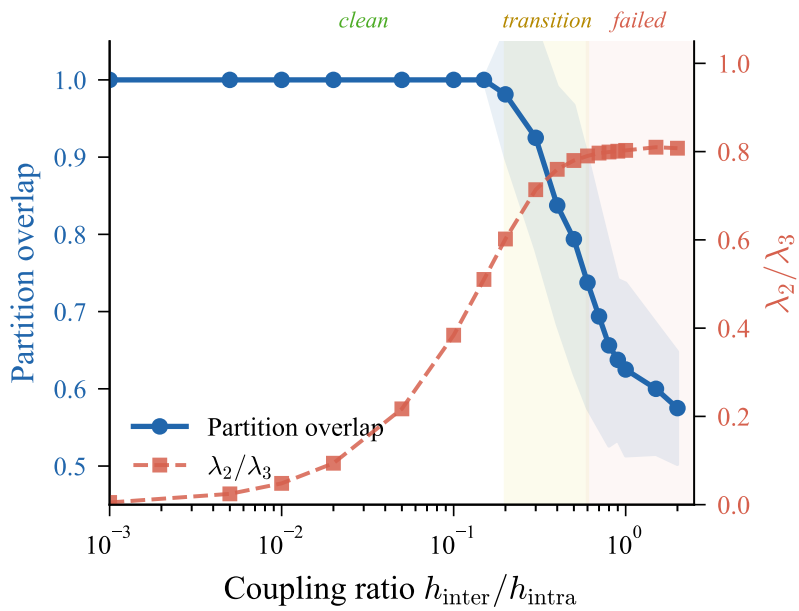

FIGURE 2. Robustness of the Fiedler partition as a function of the coupling ratio $H_{\text{inter}}/H_{\text{intra}}$. Left axis (blue): overlap between the Fiedler-predicted partition and the true block partition, averaged over 20 random Hamiltonian instances. Right axis (orange): the diagnostic ratio λ_2/λ_3, which quantifies spectral separation. In the clean regime (ratio $\lesssim 0.3$), the Fiedler partition is exact (overlap $= 1.0$) and the diagnostic is small. As inter-block coupling increases, the partition degrades through a transition regime into failure. The diagnostic λ_2/λ_3 tracks the transition faithfully, providing a built-in reliability indicator with no external oracle.

Proof sketch. Under selective dephasing (Theorem 3.1), all inter-sector coherences decay at rate γ_{inter} while all intra-sector coherences decay at rate $\gamma_{\text{intra}} \ll \gamma_{\text{inter}}$. The block partition (A, B) is therefore the first to satisfy the formation criterion $\|\rho_{AB}\|_F < \varepsilon$, for any threshold ε in the achievable range. Any competing partition $(S, S^c) \neq (A, B)$ has at least one boundary coherence decaying at the slower rate γ_{intra} and cannot form before (A, B). The Fiedler eigenvector of $L(G_H)$ identifies this partition via the Davis–Kahan theorem: when $H_{\text{inter}}/H_{\text{intra}} < \theta_c$, the sign partition of the Fiedler vector is exactly (A, B). $\square$

The threshold independence is significant: within the regime where the monitoring conditions hold and $H_{\text{inter}}/H_{\text{intra}} < \theta_c$, the branch partition is a *topologically stable* feature of the coherence graph. The partition does not depend on how sharply one draws the line between "coherent" and "decohered" — only on the community structure of the coupling graph. For the primary 4-state model used throughout this paper, $\theta_c = 2.36$, while $\|H_{\text{inter}}\|_F/\|H_{\text{intra}}\|_F = 0.0375$ — a margin of 63×. The Fiedler gap ratio $\lambda_2/\lambda_3 = 0.0375$ and the Davis–Kahan bound gives $\sin\angle(\hat{v}_2, v_2^{(0)}) \leq 0.038$, confirming that the numerical examples sit deep within the universality regime. Threshold insensitivity is confirmed directly by sweeping ε over two orders of magnitude (Test 3 in [23]), and the partition is further validated across 50 random Hamiltonian instances per coupling ratio, 20 random pure initial states, and 10 random mixed initial states (Tests 1 and 4 in [23]).

2.7. **Basis selection.** The derivation holds in any basis, but different bases yield different graphs. The physically meaningful basis is the pointer basis selected by einselection [4]: the eigenbasis of the system operator in the system-environment coupling. Given $H_{\text{int}} = S_{\text{op}} \otimes E_{\text{op}}$, the pointer states are the eigenstates of S_{op}.

That this is the correct basis follows from a simple observation. Under Lindblad dynamics, the Hamiltonian contributes identically zero to the rate of purity change:

$$\frac{d}{dt}\mathrm{Tr}(\rho^2)\bigg|_H = -\frac{i}{\hbar}\mathrm{Tr}\big([H, \rho^2]\big) = 0, \tag{4}$$

by cyclicity of the trace. The entire purity loss is due to the dissipator. The basis that minimises purity loss — Zurek's predictability sieve [4] — selects eigenstates of the decoherence operator, which are the pointer states. While the coherence graph construction is formally basis-dependent, the physically relevant basis is selected dynamically by the system-environment interaction, not chosen by the theorist.

3. Branch Formation

Branching is a subsystem phenomenon. The total state of an isolated system evolves unitarily and does not branch. Branching is what happens to a subsystem: the reduced state $\rho_{SA} = \mathrm{Tr}_E(\rho_{SAE})$, obtained by tracing over the environment, is an open system whose coherence graph fragments.

3.1. **The measurement Hamiltonian.** Consider $S \otimes A \otimes E$ with total Hamiltonian:

$$H_{\text{total}} = H_{SA} \otimes I_E + I_{SA} \otimes H_E + I_S \otimes \tilde{A} \otimes B, \tag{5}$$

where:

$$H_{SA} = H_S \otimes I_A + I_S \otimes H_A + \sum_k |s_k\rangle\langle s_k| \otimes F_k. \tag{6}$$

The interaction $\sum_k |s_k\rangle\langle s_k| \otimes F_k$ is the von Neumann measurement form: diagonal in the system index. The environment couples to the apparatus through $I_S \otimes \tilde{A} \otimes B$; the I_S factor means the environment has no direct channel to S.

The matrix elements of H_{SA} in the product basis $|s_k, a_j\rangle$ split by type:

$$\langle s_k, a_j|H_{SA}|s_k, a_m\rangle = \langle a_j|H_A + F_k|a_m\rangle \quad (k = l), \tag{7}$$

$$\langle s_k, a_j|H_{SA}|s_l, a_j\rangle = \langle s_k|H_S|s_l\rangle \quad (k \neq l), \tag{8}$$

$$\langle s_k, a_j|H_{SA}|s_l, a_m\rangle = 0 \quad (k \neq l,\, j \neq m). \tag{9}$$

The coupling graph G_H has block-structured topology: dense intra-sector edges from $H_A + F_k$, sparse inter-sector edges from H_S.

3.2. **Selective dephasing.** Since H_{SA} is block-diagonal in the system index, its eigenstates are $|s_k\rangle \otimes |a_j^{(k)}\rangle$, where $|a_j^{(k)}\rangle$ are eigenstates of $H_A + F_k$. The SA coupling operator $A = I_S \otimes \tilde{A}$ is block-diagonal in the system index:

$$\langle s_k, a_j^{(k)}|A|s_l, a_m^{(l)}\rangle = \delta_{kl}\, \langle a_j^{(k)}|\tilde{A}|a_m^{(k)}\rangle. \tag{10}$$

Under the Born–Markov-secular approximation [7], the pure dephasing rate between eigenstates $|n\rangle$ and $|m\rangle$ of H_{SA} is $\gamma_{nm}^{(\text{pure})} = \tilde{\gamma}(0)\, |A_{nn} - A_{mm}|^2$. Define the *environmental address*:

$$\tilde{a}_k(j) := \langle a_j^{(k)}|\tilde{A}|a_j^{(k)}\rangle. \tag{11}$$

Definition 3.1 (Monitoring condition)**.** The environment monitors the pointer variable if there exist sector labels $\{\bar{a}_k\}$, an energy cutoff J, and constants $\Delta, \sigma > 0$ such that for all low-energy eigenstates $j \leq J$ and all sectors k:

$$|\tilde{a}_k(j) - \bar{a}_k| \leq \sigma, \qquad |\bar{a}_k - \bar{a}_l| \geq \Delta \ \text{ for } k \neq l, \qquad \Delta \gg \sigma. \tag{12}$$

Theorem 3.1 (Selective dephasing)**.** Under premises:

(M1) $H_{\text{int}}(SA)$ is of von Neumann measurement type;
(M2) the environment couples to the apparatus only: $H_{\text{int}}(AE) = I_S \otimes \tilde{A} \otimes B$;

(M3) the monitoring condition holds with separation Δ and spread σ;
(M4) pure dephasing dominates relaxation;

and within the Born–Markov-secular approximation [7], the dephasing rates for low-energy eigenstates satisfy:

$$\frac{\gamma_{\text{inter}}}{\gamma_{\text{intra}}} \geq \left(\frac{\Delta - 2\sigma}{2\sigma}\right)^2 \gg 1. \tag{13}$$

Proof. Inter-sector ($k \neq l$): $|\tilde{a}_k(j) - \tilde{a}_l(m)| \geq \Delta - 2\sigma$. Intra-sector ($k = l$): $|\tilde{a}_k(j) - \tilde{a}_k(j')| \leq 2\sigma$. Squaring and multiplying by $\tilde{\gamma}(0)$ gives $\gamma_{\text{inter}} \geq \tilde{\gamma}(0)(\Delta - 2\sigma)^2$ and $\gamma_{\text{intra}} \leq \tilde{\gamma}(0)(2\sigma)^2$. □

Condition M4 is not an additional assumption for the inter-sector rates that drive branch formation: since the pointer states are eigenstates of the coupling operator, the transition matrix elements A_{nm} between pointer states in different sectors vanish identically (A is block-diagonal), while the diagonal differences $|A_{nn} - A_{mm}|$ are maximal by the monitoring condition. Thus $\gamma_{\text{inter}}^{(\text{relax})} = 0$ exactly for the Lindbladian generated by $\tilde{A}$.

Premises M1–M3 are *sufficient conditions*, not foundational assumptions. The Fiedler eigenvector discovers the partition as a spectral invariant; M1–M3 are then checkable predictions: given the Fiedler-discovered partition, one verifies that the Hamiltonian has the required structure. If the conditions fail, the definition reports that this Hamiltonian does not produce branches. The selective dephasing rate ratio is validated numerically using exact unitary evolution: the predicted lower bound of 64× is confirmed, with inter-sector coherences decaying at $\gamma_{\text{inter}} \approx 4.1$ (Test 5 in [23]).

3.3. **The formation theorem.** Write $\rho(t)$ in block form with respect to the partition (A, B): let $M_{AB}(t) \equiv \rho_{\text{inter}}(t)$ denote the off-diagonal block, and define $x(t) := \|M_{AB}(t)\|_F$ (inter-sector coherence), $y(t) := \|M_{AA}(t)\|_F$ (intra-sector coherence), $C := 2\|H_{AB}\|_F/\hbar$.

Theorem 3.2 (Branch formation)**.** Let G_H be block-structured with partition (A, B). Suppose: (i) $\gamma_{\text{inter}} > \gamma_{\text{intra}}$ (Theorem 3.1); (ii) $C < \varepsilon\,\gamma_{\text{inter}}$; (iii) $y_0(\gamma_{\text{inter}} - \gamma_{\text{intra}}) > Cx_0$. Then at

$$t^* = \frac{1}{\gamma_{\text{inter}}} \log\left(\frac{x_0}{\varepsilon - C/\gamma_{\text{inter}}}\right), \tag{14}$$

inter-sector coherences satisfy $x(t^*) < \varepsilon$ while intra-sector coherences remain nonzero: $y(t^*) > 0$. Figure 3 illustrates this separation numerically, and Figure 4 shows the corresponding spectral signature: the

algebraic connectivity λ_2 of $G_\rho(t)$ drops to zero as the graph fragments from one connected component into two. The predicted formation time is validated against exact unitary simulation: for the primary model, $t^*_{\text{pred}} = 0.95$ versus $t^*_{\text{obs}} = 0.94$ (Test 6 in [23]).

Proof. Step 1. Gronwall bound: $x(t) \leq x_0 e^{-\gamma_{\text{inter}} t} + C/\gamma_{\text{inter}}$. Setting this to ε gives t^*.

Step 2. Duhamel bound on the intra-sector semigroup:

$$y(t^*) \geq e^{-\gamma_{\text{intra}} t^*} y_0 - \frac{C x_0 (e^{-\gamma_{\text{intra}} t^*} - e^{-\gamma_{\text{inter}} t^*})}{\gamma_{\text{inter}} - \gamma_{\text{intra}}}.$$

Since $\gamma_{\text{intra}}\, t^*$ is small when $\gamma_{\text{inter}} \gg \gamma_{\text{intra}}$, $e^{-\gamma_{\text{intra}} t^*}$ stays near 1 while $e^{-\gamma_{\text{inter}} t^*}$ is small. Condition (iii) ensures $y(t^*) > 0$. □

Remark 3.1 (Why branches persist). For pure dephasing with any Hamiltonian H, the maximally mixed state $\sigma = (1/d)I$ is an exact faithful stationary state of the Lindbladian $\mathcal{L}$, which satisfies KMS detailed balance and is self-adjoint in the KMS inner product. Self-adjointness guarantees a real, non-positive spectrum with a spectral gap $\Delta > 0$ separating the zero eigenspace (sector-diagonal states) from the first excited eigenspace. Any inter-sector coherence regenerated by H decays at rate $\geq \Delta$. Branch persistence is a permanent consequence of the Lindbladian's spectral structure, not a transient property of the formation time. For finite-dimensional systems, Poincaré recurrences can temporarily restore inter-sector coherence, but the recurrence time scales exponentially in the number of environmental degrees of freedom N_E. For macroscopic environments ($N_E \sim 10^{23}$), recurrences are unphysical on any accessible timescale.

4. Riedel's Criteria

Riedel [9] identifies four criteria that a satisfactory branch definition must meet. We show that the definition satisfies all of them.

4.1. **Criterion 1: Tree structure.**

Theorem 4.1 (Sector-level tree structure)**.** Under the conditions of Theorem 3.2, for all $t \geq t^*$:

(i) $\|M_{AB}(t)\|_F < 2\varepsilon$. Sectors once formed remain formed.
(ii) No connected component of $G_\rho(t)$ at threshold 2ε can span both sectors.

The fragmentation forms a forward-in-time tree: components split but never merge.

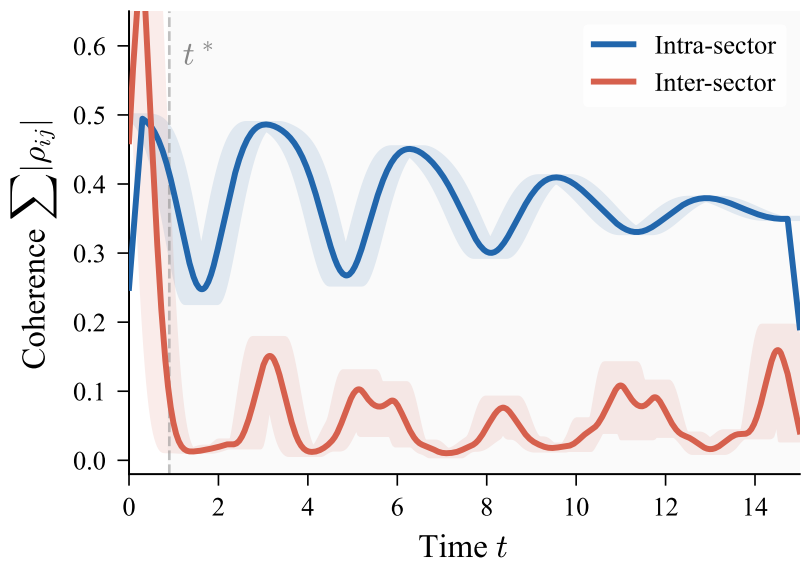

FIGURE 3. Selective dephasing and branch formation in a spin-1/2 system coupled to a 6-qubit environment via a von Neumann measurement Hamiltonian (dimension $4 \times 64 = 256$). Blue: intra-sector coherences $\sum|\rho_{ij}|$ within each Fiedler sector. Orange: inter-sector coherences between sectors. Solid lines show rolling averages; shaded bands show the oscillation range. The formation time t^* (vertical dashed) marks when the inter-sector coherence first crosses the threshold ε. After t^*, the coherence graph is fragmented: inter-sector coherences remain at a residual level while intra-sector coherences persist.

Proof. (i) Apply the Gronwall bound starting at t^* with $x(t^*) < \varepsilon$:

$$x(t) \leq \varepsilon\, e^{-\gamma_{\text{inter}}(t-t^*)} + C/\gamma_{\text{inter}} < \varepsilon + \varepsilon = 2\varepsilon,$$

using condition (ii) of Theorem 3.2: $C/\gamma_{\text{inter}} < \varepsilon$.

(ii) Suppose a component D of $G_\rho(t_2)$ contains $m \in A \cap D$ and $n \in B \cap D$. Since D is connected at threshold 2ε, some path crosses the A/B boundary at an edge (p, q) with $|\rho_{pq}(t_2)| \geq 2\varepsilon$. But $|\rho_{pq}(t_2)| \leq \|M_{AB}(t_2)\|_F < 2\varepsilon$ from part (i). Contradiction. □

The full tree is built inductively. Each level requires a physical decoherence event with its own selective dephasing condition. Each sub-fragmentation is irreversible under the same Gronwall bound (verified numerically in Test 7 of [23]). Within a branch, the dynamics remains fully quantum: intra-sector coherences are maintained, and

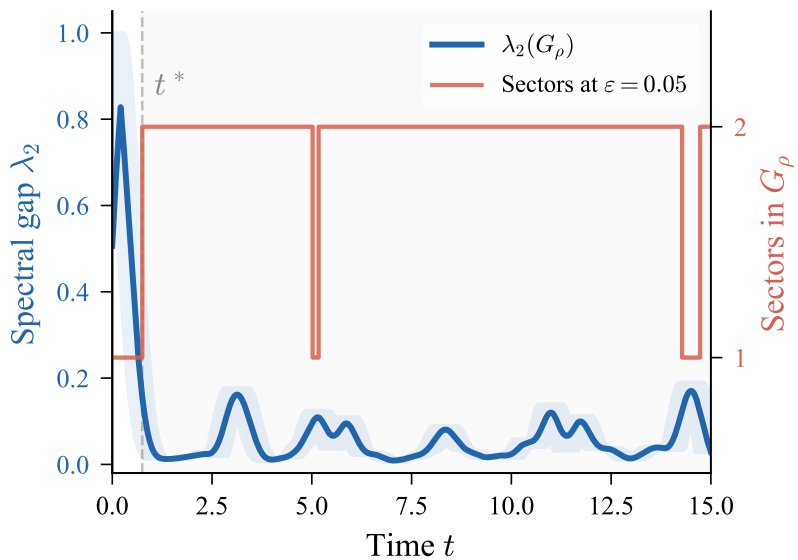

FIGURE 4. Branch formation as a spectral transition. Blue: the Fiedler eigenvalue λ_2 of the coherence-graph Laplacian $L(G_\rho(t))$, which measures algebraic connectivity. At $t = 0$ the graph is fully connected ($\lambda_2 = 1$); decoherence drives λ_2 toward zero as inter-sector edges weaken. Orange: the number of connected components (sectors) of $G_\rho(t)$ at threshold $\varepsilon = 0.05$. The topological transition $1 \to 2$ coincides with the spectral gap closing. Brief reversions to 1 sector correspond to coherence revivals visible as peaks in λ_2. Same model as Figures 1 and 3.

sub-fragmentation occurs only when new physics provides a new decoherence channel.

4.2. **Criterion 2: Quasiclassical eigenstates.** After fragmentation, the state within sector k is approximately $\rho_k \approx |s_k\rangle\langle s_k| \otimes \rho_A^{(k)}$.

Proposition 4.1 (Pointer variance bound)**.** Let $S_{\mathrm{op}} = \sum_k s_k\, |s_k\rangle\langle s_k|$ be the pointer observable and let ρ_k be the normalised branch state in sector k after formation (Theorem 3.2). Then

$$\mathrm{Var}_{S_{\mathrm{op}}}(\rho_k/p_k) \;\leq\; 4\|S_{\mathrm{op}}\|_{\mathrm{op}}^2\, \frac{\sqrt{2d_k d_{\bar{k}}}\,\varepsilon}{p_k}, \tag{15}$$

where $p_k = \mathrm{Tr}(\rho_k)$ is the branch weight, d_k is the sector dimension, and $d_{\bar{k}} = d - d_k$.

Proof. Write $\rho_k = \rho_k^{(0)} + \delta\rho_k$, where $\rho_k^{(0)} = |s_k\rangle\langle s_k| \otimes \rho_A^{(k)}$ is the exactly sector-diagonal part. Since $S_{\mathrm{op}}\,\rho_k^{(0)} = s_k\,\rho_k^{(0)}$, we have $\mathrm{Var}_{S_{\mathrm{op}}}(\rho_k^{(0)}/p_k) = 0$ exactly. By the branch definition, $|(\delta\rho_k)_{ij}| < \varepsilon$ for all inter-sector pairs, giving $\|\delta\rho_k\|_1 \leq \sqrt{2d_k d_{\bar{k}}}\,\varepsilon$. Standard perturbation of variance gives the bound. □

For macroscopic systems the threshold ε is exponentially small in the number of environmental degrees of freedom, while d/p_k is at most polynomial. Branches are sharp eigenstates of the pointer observable (validated in Test 8 of [23]).

4.3. **Criterion 3: Effective collapse.**

Proposition 4.2 (Expectation value decoupling)**.** Under the conditions of Theorem 3.2, let $\rho = \sum_k p_k\,\rho_k + \rho_{\mathrm{cross}}$ be the decomposition into branch states and cross-branch coherences. For any observable O:

$$\left|\langle O\rangle - \sum_k p_k\,\mathrm{Tr}(O\,\rho_k/p_k)\right| \leq \|O\|_{\mathrm{op}}\,\|\rho_{\mathrm{cross}}\|_1. \tag{16}$$

After formation, $\|\rho_{\mathrm{cross}}\|_1 \leq \sqrt{d}\,\varepsilon$.

The dynamical counterpart is the flow current bound (3): amplitude ceases to flow between branches. Each branch evolves autonomously, to within a residual controlled by $C/\gamma_{\mathrm{inter}}$ (validated in Test 9 of [23]).

4.4. **Criterion 4: Bounded entanglement.**

We establish two results: a Fannes bound showing that entanglement entropy across the system-apparatus cut vanishes as $\varepsilon \to 0$, and an area-law [12] for coherence crossing any spatial bipartition of the apparatus.

Proposition 4.3 (Entanglement bound)**.** Let $\rho_k = \rho_k^{(0)} + \delta\rho_k$ be a branch state with $\|\delta\rho_k\|_1 \leq \sqrt{2d_k d_{\bar{k}}}\,\varepsilon$ (Proposition 4.1). Setting $\eta := \sqrt{2d_k d_{\bar{k}}}\,\varepsilon/p_k$, Fannes' inequality [15] gives:

$$S(\mathrm{Tr}_A\,\rho_k/p_k) \leq \eta\,\log(d_S) + H_{\mathrm{bin}}(\eta). \tag{17}$$

For $\varepsilon \to 0$, entanglement across the $S|A$ cut vanishes.

For quantum correlations within the apparatus across a spatial bipartition, the coherence graph provides a direct bound.

Lemma 4.1 (Decoherence-induced coherence decay)**.** Let $H_A + F_k$ be a local Hamiltonian on a lattice with interaction range r [14], coordination number z, and maximum coupling strength $h = \max_{ij}|H_{ij}|$. Under Lindblad evolution with intra-sector dephasing rate γ_{intra}, define

the damping parameter $\lambda = 2zh/\gamma_{\text{intra}}$. If $\lambda < 1$, the quasi-stationary off-diagonal elements satisfy:

$$|\rho_{ij}|_{\text{ss}} \leq \rho_{\max}\, \mu^{d(i,j)/r}, \tag{18}$$

where $\rho_{\max} = \max_i \rho_{ii}$, $d(i,j)$ is the graph distance, and $\mu = (1 - \sqrt{1-\lambda^2})/\lambda < 1$.

Proof. In the quasi-stationary regime, the off-diagonal evolution gives:

$$\gamma_{\text{intra}}\, |\rho_{ij}| \leq \sum_k \big(|H_{ik}|\, |\rho_{kj}| + |\rho_{ik}|\, |H_{kj}|\big). \tag{19}$$

Locality constrains the sum to z neighbours. Defining $M(d) = \max_{d(i,j)=d} |\rho_{ij}|$, this yields the recurrence $M(d) \leq (\lambda/2)[M(d-1) + M(d+1)]$. The comparison function $f(d) = \rho_{\max}\, \mu^d$ satisfies the recurrence with equality, where μ is the smaller root of $(\lambda/2)\mu^2 - \mu + (\lambda/2) = 0$. A discrete maximum principle argument establishes $M(d) \leq f(d)$. □

The condition $\lambda < 1$ follows from macroscopic environment size: $\lambda = 2zh/\gamma_{\text{intra}} \sim 1/N_E$, so for any finite local coupling h, $\lambda \ll 1$ for macroscopic environments.

Proposition 4.4 (Area-law from decoherence)**.** Under premises M1–M4, with a local Hamiltonian and macroscopic environment ($\lambda < 1$), the total coherence crossing any spatial bipartition (R, R^c) of the apparatus satisfies:

$$C_{\text{cross}}(R, R^c) := \sum_{i \in R,\, j \in R^c} |\rho_{ij}| \;\leq\; |\partial R|\, \rho_{\max}\, \frac{\mu}{1-\mu}. \tag{20}$$

This scales with the boundary area $|\partial R|$, not the volume.

Proof. Partition crossing pairs by graph distance from the boundary. At distance d, there are at most $c_D\, |\partial R|\, d^{D-1}$ pairs. By Lemma 4.1, each contributes at most $\rho_{\max}\, \mu^d$. The series converges for $\mu < 1$. □

Remark 4.1 (Crossing coherence vs. entanglement entropy)*.* C_{cross} directly bounds the observable consequence of cross-boundary quantum correlations for every local observable: $|\text{Tr}(O\, \rho_{\text{cross}})| \leq \|O\|_{\text{op}}\, C_{\text{cross}}$. An area-law for C_{cross} implies that branches are operationally local: no local experiment within a region R can detect more than $O(|\partial R|)$ worth of quantum correlation with the complement. The exponential decay of coherence with graph distance is validated numerically in Test 10 of [23].

5. Known Results

The theorems above are proved from their stated premises. We now verify that the definition reproduces the correct physics in standard experiments. These examples demonstrate the definitions; they do not carry the weight of the results. All numerical simulations use exact unitary evolution followed by partial trace — no Lindblad equation or Markov approximation is employed in the numerics. The companion suite [23] extends these demonstrations with systematic validation: random Hamiltonian ensembles mapping the Fiedler partition's failure boundary (Test 1), initial-state sweeps confirming partition universality across 30 random states (Test 4), monitoring-structure stress tests showing that branching degrades when M1–M3 are violated (Test 15), a multi-branch ($k = 3$) scenario demonstrating sequential bisection (Test 2), and an environment scaling study confirming that λ_2 is independent of d_E while formation quality improves monotonically from $d_E = 4$ to $d_E = 256$ (Test 14). All simulations are limited to total dimension ≤ 1024; extension to macroscopic dimensions is untested.

5.1. **Stern–Gerlach.** A spin-1/2 particle enters a Stern–Gerlach apparatus in state $|\psi(0)\rangle = (\alpha|\uparrow\rangle + \beta|\downarrow\rangle) \otimes |p_0\rangle$. The interaction $H_{\text{int}} = \mu(\partial B/\partial z)\,\sigma_z \otimes \hat{z}$ is of von Neumann type. The coupling graph G_H has block structure. As the beams separate, the monitoring condition is satisfied: sector separation Δ grows with beam displacement while intra-sector spread σ remains at wavepacket width. By Theorem 3.1, selective dephasing holds once $\Delta \gg \sigma$. By Theorem 3.2, the coherence graph fragments into two branches with weights $|\alpha|^2$ and $|\beta|^2$ (Figure 5; Test 11 in [23]). No collapse postulate is required.

5.2. **Double-slit.** The intensity at screen position x under partial decoherence is:

$$I(x,t) = I_0\big[1 + |\rho_{LR}(t)|\cos(2\pi dx/\lambda D)\big], \tag{21}$$

where the fringe visibility $V = |\rho_{LR}(t)|$ is the coherence graph edge weight between paths L and R. The quantum-to-classical transition is the continuous suppression of a single edge in $G_\rho(t)$ (Test 12 in [23]).

5.3. **Bell correlations.** For the partially dephased singlet with normalised coherence weight V:

$$S_{\max}(t) = 2\sqrt{1 + V(t)^2}, \tag{22}$$

from the Horodecki criterion [8]. Bell violation persists for all $V > 0$. Entanglement is a global coherence structure; nonlocal correlations persist until the coherence weight is fully suppressed (Test 13 in [23]).

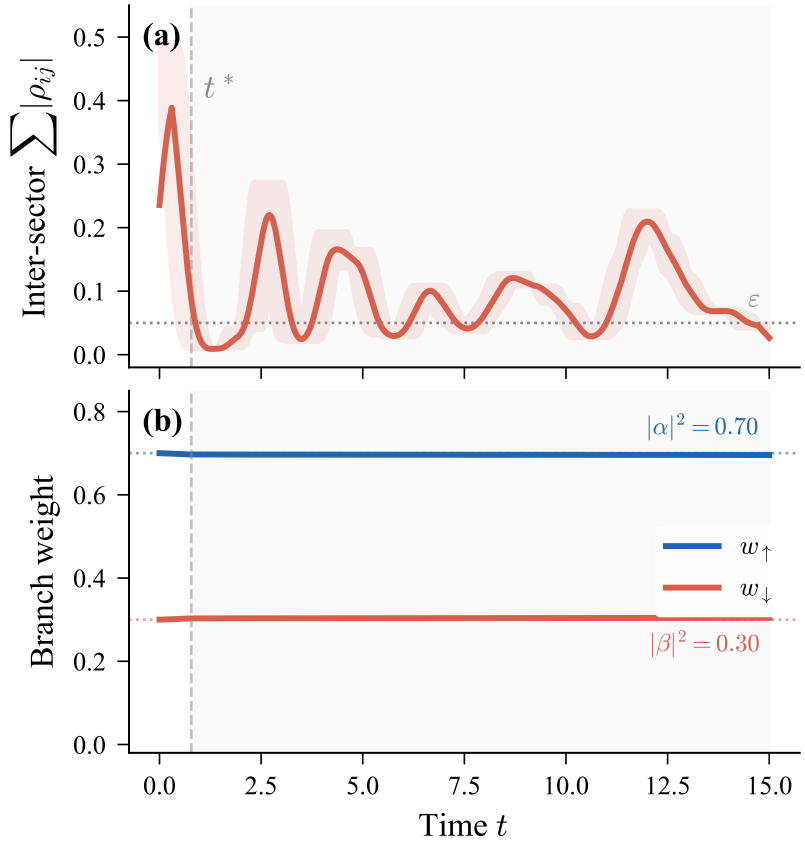

FIGURE 5. Stern–Gerlach pipeline: numerical verification with unequal branch weights ($|\alpha|^2 = 0.70$, $|\beta|^2 = 0.30$). Spin-1/2 system with 4-site apparatus and 6-qubit environment (total dimension $8 \times 64 = 512$). (a) Inter-sector coherence $\sum |\rho_{ij}|$ decays sharply at the formation time t^* and remains at a residual level (shaded band shows oscillation range). (b) Branch weights stabilise at the Born-rule values 0.70 and 0.30 (dotted lines), confirming that the definition recovers the correct probabilities.

6. Relationship to Existing Work

Riedel [9] evaluates two complexity-based proposals: Weingarten's minimisation of expected complexity plus entropy [10], and Taylor & McCulloch's interference-distinguishability gap [11]. Table 1 summarises the comparison.

TABLE 1. Comparison of branch definition proposals. All three require structural input beyond the raw wavefunction.

Feature	**Weingarten**	**Taylor & McCulloch**	**This paper**
Unique decomposition	Yes (by minimisation)	No (multiple "good" decompositions)	Yes (Fiedler partition)
Computable	No (complexity is generically infeasible)	No (complexity is generically infeasible)	Yes (spectral, $O(n^2)$)
Tree structure	Heuristic (2nd law of complexity)	Heuristic (2nd law of complexity)	Proved (Gronwall bound)
Bounded entanglement	Argued (length scale of b)	Not addressed	Proved (Lemma 4.1)
Relativistic	Yes (Lorentz covariant)	No	No
Required input	Elementary operations, vacuum Ω, parameter b	Lattice, elementary gates, error ε	Monitoring structure (A4)

All three proposals require structural input beyond the raw wavefunction. Weingarten requires elementary operations, a vacuum state, and a free parameter b controlling the branch count. Taylor & McCulloch require a lattice, elementary gates, and an error tolerance. This paper requires the monitoring structure (Axiom A4). None achieves a fully input-free definition. A notable distinction is that the coherence threshold ε does not affect the branch partition (Theorem 2.1), whereas Weingarten's b directly controls how many branches appear.

Both complexity-based proposals invoke the second law of quantum complexity to argue that branches persist. This paper's stability argument is structural: the KMS spectral gap of the Lindbladian guarantees that any regenerated inter-sector coherence decays at a rate set by the spectral gap (Remark 3.1).

Weingarten addresses Lorentz covariance; this paper does not. This is a genuine limitation of the present approach, which requires a preferred time slicing through the Hamiltonian formulation.

Relationship to decoherent histories. The decoherent (consistent) histories formalism [18, 19] provides an established framework for discussing branch-like structures in quantum mechanics. In that formalism, one proposes a set of coarse-grained histories and checks whether the decoherence functional $D(\alpha, \beta) = \mathrm{Tr}(C_\alpha \, \rho \, C_\beta^\dagger)$ is approximately diagonal; if so, the histories are said to decohere and probabilities can be assigned. The coherence graph construction differs in three respects. First, the decoherent histories framework requires as input a choice of coarse-graining (which histories to check); the coherence graph construction outputs the partition from the Hamiltonian's spectral structure without a prior proposal. Second, the decoherence functional is defined for temporal sequences of projectors (histories), while the coherence graph operates on the instantaneous reduced density matrix at a single time. Third, the decoherent histories framework can produce multiple incompatible sets of consistent histories (the "set selection problem"); the Fiedler partition is unique. However, the two frameworks are related: when the decoherence functional is approximately diagonal for the pointer-basis projectors, the corresponding off-diagonal density matrix elements $\rho_{ij}(t)$ are small, which is exactly the condition for fragmentation of $G_\rho(t)$. The coherence graph construction can thus be viewed as providing an algorithmic procedure for identifying the coarse-graining under which histories decohere, rather than as a replacement for the decoherent histories framework itself.

Why this definition? Among the available proposals, the coherence graph construction is distinguished by three properties: (i) it is the unique definition that produces a partition as the output of a polynomial-time spectral computation with no optimisation, variational freedom, or free parameters beyond the threshold ε (which does not affect the partition); (ii) it comes equipped with a built-in diagnostic (λ_2/λ_3) that reports when the definition does not apply, rather than producing a spurious partition; and (iii) all four of Riedel's criteria are proved, not assumed or argued heuristically. A natural algebraic alternative would be to define branches as the sectors of the decoherence-free subalgebra of the Lindbladian. This is mathematically well-defined and requires no threshold. However, the decoherence-free subalgebra characterises the *asymptotic* invariant subspace and does not provide a procedure for discovering the partition from the Hamiltonian; it requires solving the full Lindbladian eigenvalue problem, which presupposes the

open-system description that the Fiedler construction is designed to analyse. The coherence graph approach discovers the partition from $|H_{ij}|$ alone, before any dynamical evolution.

6.1. **Convergent programmes.** The branching problem intersects several active research programmes that approach the same set of questions from different directions. We briefly describe the connections.

Wallace and the status of branches. Wallace [3] argues that branches are emergent, higher-level structures — "real patterns" in Dennett's sense — and that demanding a precise mathematical definition reflects a category error. The coherence graph construction can be read as partial vindication: branches *are* emergent from the dynamics, as Wallace claims, but a precise definition is nonetheless available. The fragmentation of $G_\rho(t)$ makes the informal notion of a "real pattern" into a concrete mathematical object — connected components of a weighted graph — without sacrificing the emergent character that Wallace emphasises. The branch partition is not put in by hand; it is read off from the decoherence dynamics. Where the present work departs from Wallace is in insisting that the monitoring structure is a necessary input: branches are emergent given the physical decomposition, not emergent from the wavefunction alone.

Quantum Darwinism. Zurek's Quantum Darwinism programme [13] identifies the redundant encoding of pointer-state information in the environment as the mechanism by which classical objectivity emerges. The coherence graph construction describes the same physical process in different mathematical language. When the environment selectively dephases inter-sector coherences while preserving intra-sector coherences, the pointer-state information becomes redundantly encoded in each environmental fragment — this is the condition that Quantum Darwinism tracks via the mutual information between the system and environment fragments, and it is simultaneously the condition under which $G_\rho(t)$ fragments into disconnected components. The two frameworks thus converge on the same physical transition (the onset of classicality through environmental monitoring) but characterise it differently: Quantum Darwinism through information-theoretic redundancy, the coherence graph through topological fragmentation. A precise mapping between the redundancy plateau of $I(S : F_k)$ and the Fiedler gap λ_2 of $G_\rho(t)$ would strengthen both programmes and is a natural target for future work.

Quantum mereology. Carroll and Singh [16] address the problem upstream of branching: how to factorise Hilbert space into subsystems that exhibit quasi-classical dynamics. Their programme seeks

decompositions that minimise entanglement production or maximise classicality, starting from H alone. The present work is complementary: it takes the subsystem decomposition as given (Axiom A4) and constructs the branching theory downstream. If quantum mereology succeeds in deriving the preferred factorisation from the Hamiltonian, it would provide principled input to Axiom A4; conversely, the coherence graph construction provides the branching definition that the mereological programme needs to evaluate whether a candidate factorisation produces well-defined branches. The two programmes thus address different halves of the same problem, and each supplies what the other lacks.

Subsystem relativity. Zanardi, Barnum, and Knill [20] showed that the decomposition of a quantum system into subsystems is not unique: it depends on the choice of algebra of observables. Different algebras yield different tensor product structures and, consequently, different notions of entanglement. This result supports the present paper's treatment of the monitoring structure as a necessary physical input rather than a derivable quantity. The non-uniqueness of subsystem decomposition is precisely *why* the monitoring structure cannot be eliminated: specifying which algebra of observables is physically realised is additional information beyond H and $|\psi\rangle$.

Branch counting and probability. The branching problem connects to foundational questions about probability in Everettian quantum mechanics. Araújo [21] addresses the branch counting problem — how to assign a number of branches to a quantum state — and Sebens [22] develops a self-locating uncertainty framework in which the Born rule is derived from rational constraints on credences in a branching universe. Both programmes require as input a notion of what a branch *is*. The coherence graph definition could supply this: the connected components of $G_\rho(t)$ after fragmentation are the branches to be counted, and their weights (the partial traces $\mathrm{Tr}(\Pi_k\,\rho\,\Pi_k)$) are the quantities whose connection to probability requires justification. Whether the present definition, when fed into Araújo's counting framework or Sebens' epistemic framework, reproduces the desired results is an open question.

7. Discussion

7.1. **Summary.** This paper defines wavefunction branches as connected components of the coherence graph after environmental decoherence fragments it. The definition rests on five axioms: three from standard quantum mechanics, plus the monitoring structure as a physical input and coherence graph fragmentation as the branch definition. From

these, we derive branch formation at an explicit timescale, partition universality, and all four of Riedel's criteria. The definition is procedural: given an open-system Hamiltonian, the branch partition is the output of a polynomial-time spectral computation.

Behaviour in large-scale systems. The Fiedler gap λ_2 of the coupling graph G_H is determined by the system-apparatus Hamiltonian H_{SA} and is independent of the environment dimension d_E. For measurement-type Hamiltonians, λ_2 is set by the inter-sector coupling $\|H_S\|$ relative to the intra-sector coupling $\|H_A + F_k\|$. To see why: when $H_{\text{inter}} = 0$, the Laplacian $L(G_H)$ decomposes into two disconnected blocks and $\lambda_2 = 0$. Turning on a small inter-sector coupling H_S with $\|H_S\| \ll \|H_A\|$, standard matrix perturbation theory (applied to the graph Laplacian) gives $\lambda_2 = O(\|H_S\|^2/\|H_A\|)$, since the first-order correction vanishes by the block structure and the leading contribution comes at second order. This is confirmed numerically in Test 14 of [23], where $\lambda_2 = 2.0$ remains constant as d_E ranges from 4 to 256. What scales with d_E is the formation quality: γ_{inter} grows with the density of environment modes (through the spectral density $J(\omega)$), driving t^* down and the residual floor C/γ_{inter} to zero. In the macroscopic limit, branches form sharply and persist indefinitely, while the Fiedler partition that determines *which* branches form is fixed by H_{SA} alone.

7.2. **What this paper does not solve.** This paper provides a definition of branches. It does not solve the measurement problem. Specifically:

- *The Born rule.* Branch weights are approximately conserved after formation (a consequence of flow current suppression), but the connection of these weights to Born-rule probabilities — via Gleason's theorem, Deutsch-Wallace decision theory, or another route — is not established here.
- *Basis dependence.* The coherence graph construction is basis-dependent. The pointer basis is selected by the system-environment interaction (einselection), not by the theorist, but a full proof of basis selection including finite-time corrections remains to be given.
- *The monitoring structure.* Axiom A4 takes the system-environment split as input. Whether this input can be eliminated, and when it can be extracted from the Hamiltonian alone [16], is an open question for further work. Numerical stress tests (companion suite, Test 15) show that systematically violating each of M1–M3 causes branching to degrade: non-diagonal system-apparatus coupling (M1 violation) prevents sector formation at

coupling strength $\gtrsim 0.4$; direct system-environment coupling (M2 violation) disrupts the partition; and reducing the monitoring separation to $\Delta/\sigma \lesssim 1$ (M3 violation) eliminates selective dephasing entirely. These results support the claim that A4 is a necessary input, not merely a convenient assumption.

- *Dynamical regime.* The formation theorem is proved within the Born–Markov-secular approximation. The branch partition itself is a property of G_H and does not depend on the Markov assumption; extending the formation *proof* to non-Markovian regimes is the open problem. Numerically, all simulations in the companion suite use exact unitary evolution (which includes non-Markovian effects, recurrences, and non-secular terms). The qualitative predictions — selective dephasing, correct partition, timescale separation — hold in every case tested. Quantitatively, the Markov-predicted formation time overestimates the observed formation time by a factor of 1.5–3× (the Gronwall bound is not tight). This suggests the theorems capture the correct mechanism but the Markov approximation is not necessary for the phenomenon itself.
- *Fiedler partition regime.* The Fiedler partition is exact when the coupling graph has well-separated communities ($H_{\text{inter}}/H_{\text{intra}} \lesssim 0.3$; see Figure 2). It degrades continuously as inter-block coupling increases, and fails entirely for symmetric or near-degenerate spectra ($\lambda_2/\lambda_3 \approx 1$). The diagnostic ratio λ_2/λ_3 tracks this transition and should be reported alongside any partition result. The definition does not apply to Hamiltonians whose coupling graphs lack community structure — it reports this honestly via the diagnostic, rather than producing a spurious partition.
- *Multi-sector branching.* The Fiedler eigenvector handles $k = 2$ sectors. For $k > 2$, the partition information is carried by the first $k - 1$ Laplacian eigenvectors, and reliably recovering it is an open problem. The physically relevant case — sequential branching events, each driven by a separate decoherence channel — is a sequence of $k = 2$ bisections, each covered by the present results. A $k = 3$ numerical test (spin-1 system, 3 sectors of 2 states each) confirms that sequential Fiedler bisection correctly recovers all three sectors, both λ_2 and λ_3 drop, and branch weights converge to 1/3 within 2.2% (companion suite, Test 2).
- *Many-body scaling and finite-size effects.* The companion test suite confirms the definition for systems of total dimension up

to 512 (system-apparatus dimension 4–8, environment 64), validated across 50 random Hamiltonian instances per configuration and 4 block-size configurations including asymmetric cases (Test 1). Extension to macroscopic dimensions is open and untested. The key quantity is the Fiedler gap λ_2 of the coupling graph G_H, which is a property of the system-apparatus Hamiltonian H_{SA} and does not depend on the environment size N_E. What *does* scale with N_E is the dephasing rate: $\gamma_{\text{inter}} \propto \tilde{\gamma}(0) \propto J(\omega)$, where $J(\omega)$ is the environmental spectral density, which grows with the number of environment modes. As $N_E \to \infty$: the formation time t^* decreases (faster decoherence), the residual floor C/γ_{inter} shrinks (sharper branch formation), and the Poincaré recurrence time diverges exponentially. The branch partition itself is unchanged — it is set by $\lambda_2(G_H)$, not by N_E. What improves is the *quality* of the formation: the gap between intra-sector and inter-sector coherence widens with environment size.

- *Relativistic generalisation.* The Hamiltonian formulation requires a preferred time slicing. This limitation is shared with Taylor & McCulloch but not with Weingarten.

Acknowledgments

This work developed through an extended collaborative process with large language models. The central ideas — that branched flow phenomena suggest a geometric, flow-based approach to branch structure, and that the coherence graph construction should be derived from the von Neumann equation — originated with the author. The author acknowledges the language models as significant contributors to the development and presentation of these ideas, while taking full responsibility for the content, the claims, and any errors herein.

The numerical companion suite [23] was developed collaboratively with the assistance of large language models, including the design and implementation of the 15-test validation framework. The simulation code and numerical companion are deposited at Zenodo.

References

[1] J. von Neumann, *Mathematical Foundations of Quantum Mechanics*, Princeton University Press, 1955.

[2] H. Everett III, "Relative state formulation of quantum mechanics," Rev. Mod. Phys. **29**, 454–462 (1957).

[3] D. Wallace, *The Emergent Multiverse*, Oxford University Press, 2012.

[4] W. H. Zurek, "Decoherence, einselection, and the quantum origins of the classical," Rev. Mod. Phys. **75**, 715–775 (2003).
[5] M. Schlosshauer, *Decoherence and the Quantum-to-Classical Transition*, Springer, 2007.
[6] E. Joos et al., *Decoherence and the Appearance of a Classical World in Quantum Theory*, Springer, 2003.
[7] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press, 2002.
[8] R. Horodecki, P. Horodecki, and M. Horodecki, "Violating Bell inequality by mixed spin-1/2 states," Phys. Lett. A **200**, 340–344 (1995).
[9] C. J. Riedel, "Wavefunction branches demand a definition!," arXiv:2506.15663 [quant-ph] (2025).
[10] D. Weingarten, "Macroscopic Reality from Quantum Complexity," Found. Phys. **52**, 45 (2022), arXiv:2105.04545.
[11] J. K. Taylor and I. P. McCulloch, "Wavefunction branching: When you can't tell pure states from mixed states," Quantum **9**, 1670 (2025), arXiv:2308.04494.
[12] M. B. Hastings, "An area law for one-dimensional quantum systems," J. Stat. Mech. P08024 (2007).
[13] W. H. Zurek, "Quantum Darwinism," Nature Physics **5**, 181–188 (2009).
[14] E. H. Lieb and D. W. Robinson, "The finite group velocity of quantum spin systems," Commun. Math. Phys. **28**, 251–257 (1972).
[15] M. Fannes, "A continuity property of the entropy density for spin lattice systems," Commun. Math. Phys. **31**, 291–294 (1973).
[16] S. M. Carroll and A. Singh, "Quantum Mereology: Factorizing Hilbert Space into Subsystems with Quasi-Classical Dynamics," Phys. Rev. A **103**, 022213 (2021), arXiv:2005.12938.
[17] M. Fiedler, "Algebraic connectivity of graphs," Czech. Math. J. **23**, 298–305 (1973).
[18] R. B. Griffiths, "Consistent histories and the interpretation of quantum mechanics," J. Stat. Phys. **36**, 219–272 (1984).
[19] M. Gell-Mann and J. B. Hartle, "Classical equations for quantum systems," Phys. Rev. D **47**, 3345–3382 (1993).
[20] P. Zanardi, "Virtual quantum subsystems," Phys. Rev. Lett. **87**, 077901 (2001); P. Zanardi, D. A. Lidar, and S. Lloyd, "Quantum tensor product structures are observable induced," Phys. Rev. Lett. **92**, 060402 (2004).
[21] M. Araújo, "Branch counting in the Everett interpretation of quantum mechanics," arXiv:2301.06524 [quant-ph] (2023).
[22] C. T. Sebens and S. M. Carroll, "Self-locating uncertainty and the origin of probability in Everettian quantum mechanics," Brit. J. Phil. Sci. **69**, 25–74 (2018), arXiv:1405.7577.
[23] B. St. Laurent, "CGA Numerical Companion: Simulation results and figures for the Coherence Graph Approach," Zenodo deposit (2026).

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